

PERSONAL FINANCE · SELF-HOSTED PWA

Personal FMS

How Your System Works

A plain-language guide to the app, the plan, and the money moves that come next.

fms.argontechs.dev · Your VPS · Your data

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1 What is this app?

Personal FMS (Financial Management System) is a **private, self-hosted money app** running at `fms.argontechs.dev` on your own server. Nothing leaves your VPS — no bank syncing, no third-party cloud, no subscription. It works on your iPhone like a native app once you add it to your Home Screen.

WHAT IT IS BUILT TO DO — THREE THINGS ONLY

1. **Build a daily money-logging habit.** Log a coffee in two taps. See one honest "safe to spend" number. That's it.
2. **Make your Emergency Fund visibly grow** every time income lands — payday prompts you to transfer, and the bar moves.
3. **Kill your 18% credit card** — the single most expensive thing in your financial life right now.

Version 1 deliberately does not try to do everything. No complex budgets, no investment tracking, no net-worth dashboards. Those come later — after the habit is formed and the card is dead. An unfinished finance app helps nobody.

2 Your situation right now

Here is a clear-eyed picture of where things stand as of June 2026.

Income

Source	Amount	When
Net salary (Public Bank)	RM 5,819.50	Around the 1st–3rd
Side income A	RM 600.00	1st of the month
Side income B	RM 600.00	23rd of the month
Total monthly income	RM 7,019.50	

Monthly outgoings

Category	Item	Amount
Fixed living	Food (estimate)	RM 1,000
	Transport (estimate)	RM 450
	Digi (phone)	RM 378.60
	Electricity (est.)	RM 150
	Unifi (internet)	RM 150
	GE Critical Illness insurance	RM 350
	GE ILP (Great Wealth Enhancer) *pause this*	RM 350
	Gym	RM 199
	Subscriptions (Netflix/Spotify/YouTube)	RM 82
Debt payments	Car loan (due 22nd)	RM 904
	PTPTN (due 1st)	RM 270
	SLoan 1 (due 12th, ends Mar 2027)	RM 177.43
	SLoan 2 (due 7th, ends Oct 2026)	RM 90.83
	ShopeePayLater (due 10th)	RM 1,519.50
	Ryt PayLater (due ~22nd, Jul–Oct only)	RM 214.15
Card interest	18% revolving (~RM7,400 × 18% ÷ 12)	RM 111
Available surplus (tightest month, July)		~RM 623

THE KEY INSIGHT

You are **cash-flow positive** — roughly RM623 available every month even in the hardest month (July), rising to about **RM2,412 by month 6** once the credit card is cleared. The surplus exists. It just leaks because nothing forces a transfer and nothing rewards it. That is the exact problem this app solves.

Your debts

Debt	Balance	Rate	Priority
Credit Card	RM 7,400.76	18% APR	Kill first
Car loan	RM 73,484.67	2.44% flat	Pay minimum
PTPTN	RM 32,843.62	~1%	Minimum forever
SLoan 1	RM 177.43 × 8	~24% EIR (fixed)	Runs off Mar 2027
SLoan 2	RM 90.83 × 3	~31% EIR (fixed)	Runs off Oct 2026
ShopeePayLater	~RM 4,350	0% interest	Fixed schedule
Ryt PayLater	RM 214.15 × 4	0%	Ends Oct 2026

WHY NOT ATTACK THE SLOANS? THEY'RE EVEN HIGHER-RATE THAN THE CARD.

SLoans 1 and 2 have high EIRs on paper — but the interest is already **baked into the fixed payments**. Paying them off early barely saves anything because the loan contract pre-computes interest upfront. They finish on schedule anyway (Oct 2026 and Mar 2027). The **credit card** is the only place extra ringgit cut future interest — it's revolving, so every ringgit you pay this month saves you ~1.5 sen in interest next month.

Your liquid assets right now

Where	Amount	Note
Main bank account	RM 750	Operational account
Cash on hand	RM 272	
TNG eWallet	RM 69.95	
UOB One	RM 2.80	
Total liquid	~RM 1,094.75	Your entire liquid net worth

Your investments and insurance policies (~**RM142k in AIA**, ASNB ~RM327, GE Critical Illness) hold significant value — but they are locked or protected. Your real problem is not net worth; it is **liquidity and the 18% card**. The plan below fixes that.

THE BIG LEVER — ASK AIA ABOUT YOUR ASSURANCE ACCOUNT

Your **AIA Assurance Account holds RM63,522.97**. AIA sometimes allows low-penalty partial withdrawals from this type of account. If you can withdraw ~RM7,400 to clear the credit card outright, that is a **guaranteed, risk-free, tax-free 18% return** — better than almost any investment. Ask AIA: "Can I make a partial withdrawal from my Assurance Account, and what is the impact on coverage?" Keep the Empower Edu plan and A-LifeJoy untouched — do not surrender whole policies. (*General guidance — verify terms directly with AIA.*)

3 The plan — your money waterfall

Every ringgit you earn flows through a waterfall. Each step is only done after the one above it is satisfied. The app enforces this logic in its numbers automatically.

1 Pay all minimums and fixed obligations

Car, PTPTN, SLoans, ShopeePayLater, Ryt PayLater, insurance. These are non-negotiable. The app auto-posts them on the right date — you just need to have the money in your bank.

2 Lock in your RM1,000 starter emergency fund

Move your existing ~RM1,095 liquid cash into the RYT Bank "Emergency Fund" Save Pocket — PIDM-insured, instant access, earns ~3%. You already have this money; it just needs to be ring-fenced. The first app action is labelling what you already hold, not saving fresh. This covers the next surprise without touching the 18% card.

3 Get the credit card off 18% — try a 0% conversion first

Before throwing money at the card, ask your card issuer if they have a balance installment plan or conversion to a lower rate. The right order: (1) Your own issuer's installment/conversion plan — no new card needed. Accept only if the effective rate clearly beats 18%. (2) If not, apply for a 0% balance-transfer card on the open market (common in Malaysia — typically RM1,000–RM50,000 eligible, 12 months 0%). Either way: **stop charging daily living to this card immediately**. Route food, petrol, Digi, Unifi, gym, and subscriptions to your debit card or e-wallet instead. The app is pre-configured for this flip.

4 Pause the Great Eastern "Great Wealth Enhancer" ILP now

This RM350/month plan is currently charged to your 18% credit card — meaning you are **borrowing at 18% to fund a ~5% plan locked for 10 years**. You are about 1 month in, so almost nothing is sunk. Contact GE: (a) confirm the premium term (is it 5 or 10 years?), (b) ask about a premium holiday, paid-up option, or cooling-off cancellation, (c) ask for the current surrender value. Then stop the card charge and redirect that RM350 to the card kill. **This is not your protection plan** — your real critical illness cover is the separate GE plan (RM350, due 27th) which stays untouched.

5 Throw all remaining surplus at the credit card

With the 0% conversion active (or even just at 18%), every spare ringgit goes to the card balance — not the car, not PTPTN. Projected card-free date: **around month 6**. Once it is dead: no more ~RM111/month interest drain, and ~RM2,412/month becomes available to deploy.

6 After the card: build the full 6-month buffer

Target: **RM15,000** (about 6 months of essential costs at ~RM2,500/month) in the RYT Bank Save Pocket. Then deploy further surplus into longer-term investing — not back into the ILP by default.

Why not the car loan? Why not PTPTN?

PTPTN: ~1% effective rate. Less than what the RYT savings account earns. Never prepay it — minimum forever.

Car loan: Malaysian flat-rate hire-purchase loans (under the Hire-Purchase Act 1967) only give a small statutory rebate on early settlement. The card at 18% beats it every time. Keep minimum

payments on the car until the card is dead.

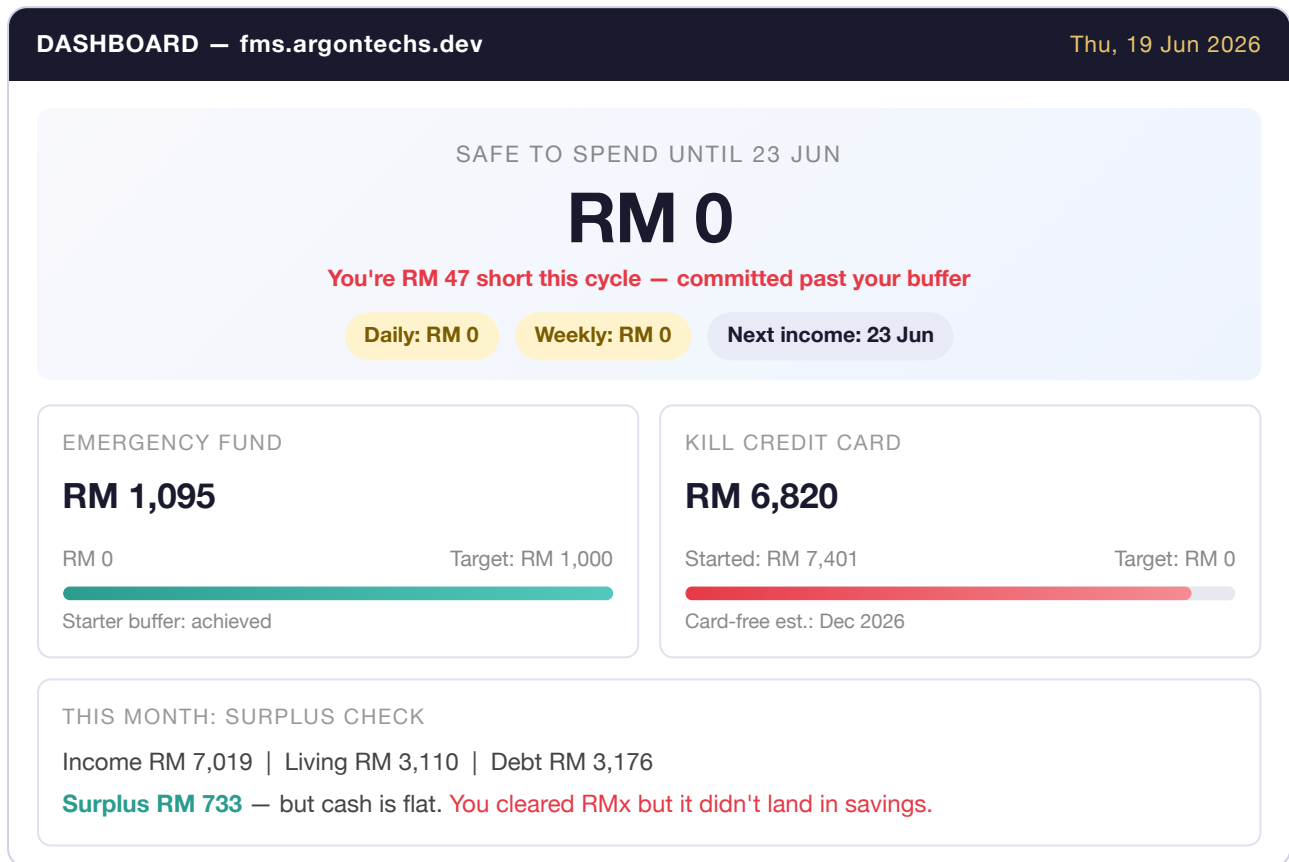
PROJECTED TIMELINE

Phase	Months	What happens
Buffer	M1–M2	Ring-fence RM1,000 EF; card gets minimum only. First habit streak.
Attack	M3–M6	Full surplus + ILP RM350 → card. Balance drops from ~RM7,400 → zero .
Sweep	M7+	SPayLater and SLoans finish on schedule. Grow EF to RM15,000.

Total card interest paid: ~~~RM444–490~~ **if no conversion; ~RM0 if the 0% BT is approved**. That ~RM490 gap is the whole point of Step 3 above.

4 The four views in the app

The app has four main screens. Here is what each one shows and how to read it.



Illustrative dashboard layout — actual values will reflect your live data.

View 1 — Safe to Spend (the hero number)

This is the most important number in the app. It answers one question: **"How much can I spend between now and my next income, without dipping below my emergency fund minimum or missing any committed payments?"**

The formula behind it:

HOW SAFE TO SPEND IS CALCULATED

Safe to Spend

- = Cash you have now
- + Any income arriving before next payday
- Every bill or debt payment due before next payday
- Your EF contribution for this cycle (if not yet done)
- RM200 hard floor (minimum always kept in account)

- **If positive:** shows the number — "Safe to spend until 23 Jun: RM347". Also shows daily and weekly breakdowns as chips.
- **If zero or negative:** shows RM0 in red with an honest explanation — "You're already RM47 short this cycle." Never shows a negative number you can somehow spend.
- It updates instantly every time you log a transaction.

View 2 — Goals

Two progress bars live here:

- **Emergency Fund:** a bar from RM0 to RM1,000 (starter), then to RM15,000 (full 6-month buffer). The bar grows when you log EF transfers on payday.
- **Kill Credit Card:** a shrinking bar from your starting balance (RM7,400.76) down to zero. It moves every time you log a card payment. Your estimated card-free date is shown here.

View 3 — Credit Card detail

This screen shows the full picture of your most expensive debt:

- Current balance and credit limit utilisation — you are currently at about **93% utilisation** (only RM587 headroom left — the card can physically decline new charges)
- Monthly interest cost (~RM111 per month right now)
- Projected card-free date, recalculated live as you log
- A recommendation to attempt the 0% balance transfer / installment conversion first
- If a 0% conversion is active: interest shown as RM0 and the countdown to the promo window closing

View 4 — This Month

A monthly summary: income received vs living expenses vs debt payments. The key nudge here is the **"leak detector"** — if your surplus is positive but your cash balance is flat, the app flags: *"You cleared RMx in surplus this month, but it hasn't landed in savings or the card."* That is the surplus-leaking problem made visible.

5 Using it day-to-day

Logging a spend — two taps

The core habit is simple. You open the app and tap the quick-log button. You enter the amount, pick a category chip (Food / Transport / Other), optionally add a note, and tap confirm. That is it. The Safe-to-Spend number updates immediately.

TWO-TAP QUICK-LOG FLOW

Open app → Tap quick-log → Type amount (e.g. 12.50) → Tap **Food** → Done.

Optional: add a note before confirming.

Offline-first — works without signal

The app is a Progressive Web App (PWA). This means it works even when you have no mobile data. Logging works offline — the entry is saved locally and synced to the server the next time you open the app with a connection. The app shell itself opens instantly, even on slow connections.

Bills and salary are auto-posted by the server on the right date — you do not need to log those manually.

Payday push notifications

On salary day (around the 1st–3rd), the 1st, and the 23rd, the app sends you a push notification:

EXAMPLE PAYDAY PROMPT

"RM600 just landed. Move RM___ to your Emergency Fund now? You're cash-flow positive — this is the surplus that usually disappears."

You tap **Transfer logged** (or Adjust / Skip). One tap moves the progress bar. Skipping is fine — it records the decision and breaks no streak.

Bill reminders

3 days before each bill is due, then 1 day before, then on the day — you get a push notification. The amount shown is always the correct one for that installment (including the declining ShopeePayLater schedule). If you update a bill in the app, the reminder amount updates automatically.

iOS — add to Home Screen to receive push

Apple requires the app to be installed to your Home Screen (not just open in Safari) before push notifications work. If you are in Safari, tap the Share button and select **Add to Home Screen**. The app detects when it is not installed and shows a one-time banner reminding you to do this. Once installed, tap "Turn on reminders" inside the app to enable push.

WHAT IF PUSH STOPS WORKING?

The app has a fallback. The dashboard always shows a "**What needs your attention**" list — overdue bills, payday savings prompts — so the habit loop closes even without push. You can also receive a weekly summary email as a backup channel.

6 First real-world steps (do these now)

The app is ready. These are the actions that make the biggest difference. Do them in this order.

#	Action	Why
1	Add the app to your iPhone Home Screen	Required for push notifications to work on iOS (Safari tab is not enough).
2	Open RYT Bank and create / confirm an "Emergency Fund" Save Pocket	PIDM-insured, instant, ~3%. Move your ~RM1,095 liquid cash here. This is already your money — you are just ring-fencing it.
3	Call your credit card issuer	Ask about a balance installment plan or installment conversion at a rate below 18%. This is the biggest single financial move available to you right now.
4	Call AIA about the Assurance Account	Ask: "Can I make a partial withdrawal of ~RM7,400 from my Assurance Account, and what is the coverage impact?" If yes, clearing the card outright is a guaranteed 18% return.
5	Contact Great Eastern — pause the Great Wealth Enhancer ILP	Ask about free-look status, premium holiday, or paid-up option. Stop the RM350 charge to the card. Redirect it to the card kill fund.
6	Switch card auto-charges to your debit card or bank	Digi, Unifi, gym, subscriptions, GE insurance — move these off the credit card. Every new charge on the card while it carries a balance costs you 18%.
7	Log your first transaction in the app	Any spend — the next coffee, petrol, lunch. Two taps. The habit starts the first time you open the app.

THE MOST IMPORTANT THING YOU CAN DO TODAY

Steps 3–5 above are not app actions — they are phone calls. They are also where the most money is. A 0% balance transfer saves you ~RM490 in card interest compared to grinding it down at 18%. Asking AIA could save the whole RM7,400 in interest immediately. The app tracks whatever you decide; it does not make those calls for you.

7 Privacy and safety

Your data stays on your server

Personal FMS is entirely self-hosted. Every transaction, balance, and debt figure is stored in a single SQLite file on your VPS at `fms.argontechs.dev`. There is no connection to any third-party service. No analytics. No bank syncing API. Nothing leaves your machine except when you make the push notification calls to Apple/Google's push servers (which contain only a notification, not your financial data).

Login security

The app uses a strong password with a server-side session. The session cookie is `httpOnly` (cannot be read by JavaScript), `Secure` (only sent over HTTPS), and bound to `fms.argontechs.dev`. Every API endpoint requires a valid session — the subscribe-for-push endpoint included. The entire site runs over HTTPS with automatic Let's Encrypt certificate renewal.

Backups

The app runs a daily automated backup of the SQLite database. Backups use the `sqlite3 .backup` command (the safe method for a live database with WAL mode enabled), compress the file, keep 14 days of local copies, and copy each backup off-server to Google Drive. A monthly restore-and-verify check confirms the backups are actually usable.

YOUR EMERGENCY FUND ITSELF IS SEPARATELY PROTECTED

The RYT Bank "Emergency Fund" Save Pocket is a real bank deposit, **PIDM-insured up to RM250,000**. The app tracks the intent and progress; the money itself lives in a protected bank account that has nothing to do with the app's server.

8 What is coming in later versions

Version 1 is deliberately focused. The following features are designed and will be added in future phases — but only after the core habit is formed. Adding them too early would increase complexity without adding value at the stage where you need to build the habit.

Phase 2 — retention and richer views (after the habit is formed)

- **Streaks and milestones** — daily logging streaks with freeze tokens ("I had nothing to log today"), milestone badges (First RM1,000 saved, card cleared, first month with positive cash, etc.)
- **Weekly check-in screen** — a 7-section guided review of your week: what was logged, where surplus went, what is due next week
- **Daily forecast chart** — a day-by-day projected cash balance chart through to the end of the month, with per-day shortfall flags
- **Per-category budget limits** — set "spend max RM1,000 on food this month" and see a coloured bar tracking toward the limit
- **Multi-debt payoff comparator** — avalanche vs. snowball, full per-debt payoff timelines
- **Passkey (biometric) login** — Face ID / Touch ID login alongside the password option

Phase 3 — accounting-grade precision

- **Full reconciliation** — compare app balance to actual bank balance, flag drift, confirm pending transactions
- **Statistical variable-spend estimation** — after 60 days of real data, the app learns your average daily spend per category instead of using a flat estimate

Deferred (net-worth / investment view)

Your AIA investment-linked accounts, ASNB holdings, and GE insurance fund values are **recorded in the system for a later net-worth view** but not tracked in v1. This is intentional — v1 focuses on cash flow and the card. Once the card is dead, a net-worth view becomes useful. The data is already structured and ready.

Holding	Recorded value	v1 status
AIA A-LifeJoy	RM 5,340.80	Deferred
AIA Assurance Account	RM 63,522.97	Deferred (but see the lever in §2)
AIA Empower Edu Plan	RM 73,032.70	Deferred
GE Critical Illness (fund value)	RM 1,426.26	Deferred — keep paying premium
ASNB (ASN Sara, Equity 5, ASM 3)	~RM 327	Deferred
Total deferred holdings		~RM 143,649

YOUR ACTUAL NET WORTH IS POSITIVE

Total holdings ~RM144k, total debt ~RM120k = **positive net worth**. The live problem is not net worth; it is liquidity (RM1,095 liquid) and the 18% card eating RM111/month. The plan is unchanged by this — fix the liquidity, kill the card, then invest.

9 In summary

WHAT THIS SYSTEM IS

A private, self-hosted money app on your own server. Its one job is to convert the RM623–RM2,412/month surplus that currently leaks into three visible outcomes: a growing Emergency Fund, a shrinking credit card, and a daily logging habit that makes the first two possible.

The math is already in your favour. The app just makes it impossible to ignore.

The three biggest moves (not in the app)

The app supports the plan. But these phone calls are where the money actually moves:

1. **Ask AIA about the Assurance Account partial withdrawal** — could clear the card instantly for a guaranteed 18% return.
2. **Ask your card issuer about a 0% installment conversion** — could save ~RM490 in future interest.
3. **Call GE and pause the ILP** — frees RM350/month and stops borrowing at 18% to fund a 5% plan.

Open the app. Log a spend. Watch the numbers move.

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